



## AQ5.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

The null space of a matrix  $A_{4 \times 3}$  is



the subspace  $W = \{x \in R^4 \mid Ax = 0\}$  of  $R^4$ .



the subspace  $W = \{x \in R^3 \mid Ax = 0\}$  of  $R^3$ .



the subspace  $W = \{x \in R^2 \mid Ax = 0\}$  of  $R^2$ .



the first column of the matrix  $A_{4 \times 3}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

the subspace  $W = \{x \in R^3 \mid Ax = 0\}$  of  $R^3$ .

1 point

Nullity of a matrix  $A_{3 \times 4}$  is



the number of independent variables in the system of linear equations  $Ax = 0$ , where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$



the number of dependent variables in the system of linear equations  $Ax = 0$ , where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

the number of independent variables in the system of linear equations  $Ax = 0$ , where  $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$



$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}.$$

Consider the system of linear equations  $Ax = 0$ , where  $x = [x_1 \ x_2 \ x_3]^T$  and  $x = [x_1 \ x_2 \ x_3]^T$  and

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Use the above information to answer questions 3, 4 and 5.

1 point

Choose the correct statement from the following options.

☐

$x_1$ ,  $x_2$  and  $x_3$  all are dependent variables.

☐

$x_2$  is the only dependent variable.

☐

$x_1$  and  $x_3$  are the dependent variables and  $x_2$  is the only independent variable.

☐

$x_1$ ,  $x_2$  and  $x_3$  all are independent variables.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$x_1$  and  $x_3$  are the dependent variables and  $x_2$  is the only independent variable.

1 point

Which one of the following vector spaces represents the null space of  $A$ ?

☐

$\{(t_1, 0, t_2) \mid t_1, t_2 \in \mathbb{R}\}$

☐

$\{(0, t, 0) \mid t \in \mathbb{R}\}$

☐

$\{(t_1, t_2, 0) \mid t_1, t_2 \in \mathbb{R}\}$

☐

$\{(0, 0, t) \mid t \in \mathbb{R}\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(0, t, 0) \mid t \in \mathbb{R}\}$

What will be the nullity of  $A$ ?

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

Find the nullity of the matrix  $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 4 & 6 \end{bmatrix}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

1 point

Let  $SS$  denote the set of solutions of the homogeneous system of linear equations  $Ax = 0$ . Which of the following statements is (are) true?

☐

If  $x_1$  and  $x_2$  are in  $SS$ , then  $x_1 - x_2$  is in  $SS$ .

☐

If  $x$  is in  $SS$ , then  $cx$  is in  $SS$ , for any  $c \in \mathbb{R}$ .

☐

If  $x_1$  and  $x_2$  are in  $SS$ , then  $\alpha x_1 + \beta x_2$  is in  $SS$ , for any  $\alpha, \beta \in \mathbb{R}$ .

☐

If  $x$  is in  $SS$ , then  $A^n x = 0$  for any  $n \in \mathbb{N} \setminus \{0\}$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $x_1$  and  $x_2$  are in  $SS$ , then  $x_1 - x_2$  is in  $SS$ .

If  $x$  is in  $SS$ , then  $cx$  is in  $SS$ , for any  $c \in \mathbb{R}$ .

If  $x_1$  and  $x_2$  are in  $SS$ , then  $\alpha x_1 + \beta x_2$  is in  $SS$ , for any  $\alpha, \beta \in \mathbb{R}$ .

If  $x$  is in  $SS$ , then  $A^n x = 0$  for any  $n \in \mathbb{N} \setminus \{0\}$ .

Find out the value of  $a$  for which the matrix  $\begin{bmatrix} 1 & 2 \\ 3 & a \end{bmatrix}$  has nullity 1.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 6

1 point

## Level 2

1 point

Consider the coefficient matrix  $A$  of the following system of linear equations:

$$x_1 + x_2 + x_4 = 0$$

$$x_2 + x_3 = 0$$

$$x_1 - x_3 + x_4 = 0$$

Which one of the following vector spaces represents the null space of  $A$ ?

☐

$\{(t_1 + t_2, t_1, t_1, t_2) \mid t_1, t_2 \in \mathbb{R}\}$

○

$$\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\} \{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$$

○

$$\{(t_1 + t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\} \{(t_1 + t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$$

○

$$\{(t_1 - t_2, -t_1, t_1, -t_2) \mid t_1, t_2 \in R\} \{(t_1 - t_2, -t_1, t_1, -t_2) \mid t_1, t_2 \in R\}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\} \{(t_1 - t_2, -t_1, t_1, t_2) \mid t_1, t_2 \in R\}$$

Find the nullity of the matrix  $AA$ , where  $A = [a_{ij}]$   $A = [a_{ij}]$  is of order  $3 \times 3 \times 3$  and  $a_{i,j} = \min\{i, j\}$   
 $a_{i,j} = \min\{i, j\}$ ,  $i, j = 1, 2, 3$ ,  $i, j = 1, 2, 3$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

1 point

Choose the correct set of options from the following.

- ☐ The nullity of a non-zero scalar matrix of order 3 must be 3.
- ☐ The nullity of a non-zero scalar matrix of order 3 must be 0.
- ☐ The nullity of a non-zero diagonal matrix of order 3 must be 3.
- ☐ The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The nullity of a non-zero scalar matrix of order 3 must be 0.

The nullity of a non-zero diagonal matrix of order 3 can be at most 2.

Consider the system  $Ax = b$   $Ax = b$ , where  $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$   $A = \begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}$ . The number of solutions of when

$$b = \begin{bmatrix} 1 \\ 0 \end{bmatrix} b = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ is}$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 0

1 point

[Check Answers](#)